

The NexMinds Data Analytics Assessment

Executive Analysis: Osmania University UG Colleges (2026 Cycle)

Candidate: Mohammed Farshan | Focus: ETL, Descriptive Analytics & Interactive Power BI Modeling

Project Objective:	Conduct an end-to-end analytical audit of undergraduate (UG) course offerings across affiliated institutions, standardizing multi-value course text into an enterprise Star Schema model and surfacing strategic insights.
Dataset Examined:	OU UG Colleges 2026 (356 Institutional Rows, 2,031 Normalized Course Offerings)
Technology Stack:	Python 3.14 (Pandas, NumPy, Matplotlib, Seaborn), Power BI Desktop (DAX), Microsoft Excel

Executive Scorecard & Key KPIs

TOTAL COLLEGES 356 355 with courses 1 without	COURSE OFFERINGS 2,031 Mean: 5.7 Median: 5.0	ENGLISH SHARE 77.6% 1,576 English courses	TOP DEGREE B.Sc 980 courses (48.3%)
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Executive Summary

This evaluation provides an authoritative analysis of undergraduate course distribution across Osmania University's affiliated colleges for the 2026 academic admissions cycle. The source dataset encompasses 356 institutional rows, which after rigorous parsing and normalization yielded 2,031 atomic course offerings across four primary degree programs: B.Sc, B.Com, B.A, and BBA.

Key strategic takeaways from the investigation demonstrate significant structural concentrations: B.Sc accounts for nearly half of all course offerings (48.25%), whereas B.Com achieves virtually universal presence across 95.2% of affiliated institutions. While English represents the prevailing language of instruction university-wide (77.6%), liberal arts (B.A) displays a profound regional language inversion, offering 1.7 times more Telugu-medium courses than English. Furthermore, course breadth is substantially skewed, with a median of only 5 courses per college while premier institutions support up to 27 distinct combinations.

Data Preparation & Structural Audit

The source dataset presented real-world data engineering challenges. Rather than storing records in normalized tables, individual college rows contained multi-value course declarations concatenated as newline-separated strings with pipe delimiters ('Degree | SubCourse | Medium').

1. Data Ingestion & Missing Value Audit

- Total Records Ingested: 356 college records across 5 initial attributes: S.No., College Code, College Name, Address/Contact/Website, and Courses.
- Null Value Audit: Zero missing values in institutional identification (College Code, College Name) or Course fields. 34 records lacked detailed website/address metadata.
- Duplicate Verification: Zero duplicate college codes were present; each code uniquely indexes an affiliated college entity.

2. The Delimiter Anomaly: Resolving the 355 vs 356 Institutional Baseline

A critical finding during ingestion was the structural discrepancy between raw college records (356) and parsed course-bearing colleges (355). College Code 1171 (Road Mystry Degree College) contained the raw course entry 'BSW\nBSW English' without pipe delimiters ('|'). Consequently, standard 3-token parsing yielded no fact records for this institution. Recognizing this anomaly is essential for maintaining dimensional integrity in enterprise Power BI reporting.

3. Transformation & Normalization Pipeline

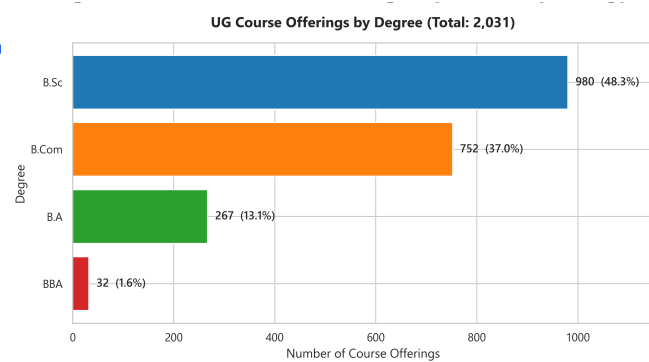
- String Unpivoting: Course blocks were split by newline ('\n') and tokenized by pipe ('|') into atomic 3-tuples.
- Degree Harmonization: Casing variations were standardized ('B.SC' -> 'B.Sc', 'B.COM' -> 'B.Com', 'B.A.' -> 'B.A').
- Medium Normalization: Text entries were trimmed and converted to Title Case ('English', 'Telugu', 'Urdu', 'Hindi').
- Data Model Output: Produced 'FactCourses' (2,031 rows x 5 columns) and 'DimCollege' (356 rows x 3 columns).

4. Power BI Star Schema Data Model

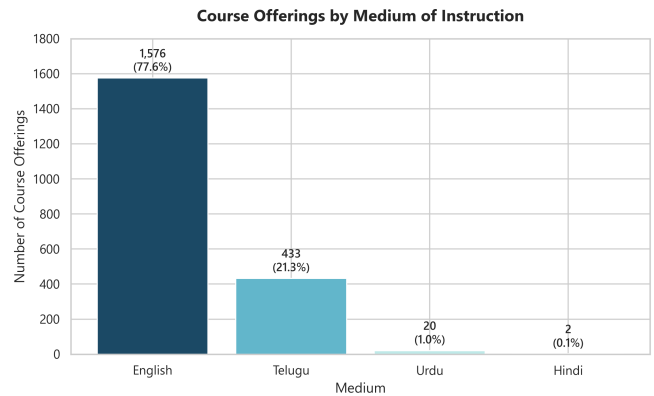
To empower performant slicing and self-service analytics, the normalized data was architected into a Star Schema data model:

- FactCourses (Fact Table): College Code (FK), College Name, Degree, SubCourse, Medium
- DimCollege (Dimension Table): College Code (PK), College Name, Address / Contact / Website
- Relationship: 1-to-Many cardinality from DimCollege[College Code] (1) to FactCourses[College Code] (*).

Table Name	Table Type	Row Count	Key Attributes
DimCollege	Dimension (1)	356	College Code (PK), College Name, Contact
FactCourses	Fact (*)	2,031	College Code (FK), Degree, SubCourse, Medium

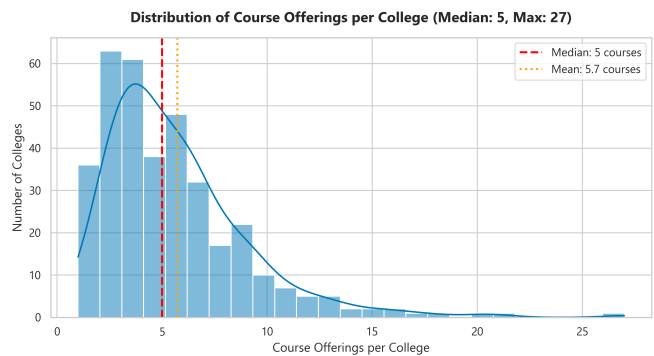
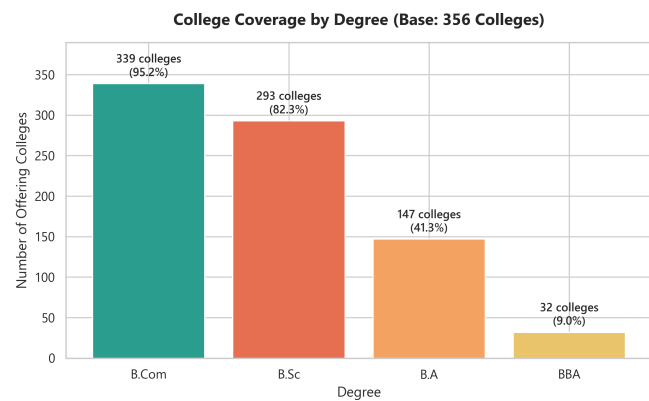


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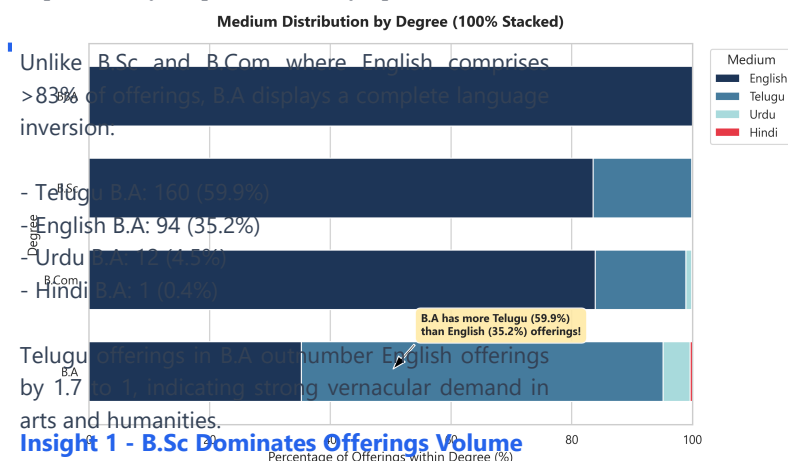
Core Distributions: Degree & Language Market Share

- **B.Sc Dominance:** B.Sc represents 980 of 2,031 course offerings (48.25%), followed by B.Com with 752 offerings (37.03%). Together, Science and Commerce constitute 85.28% of all undergraduate pathways.
- **Medium Landscape:** English is the primary medium of instruction across 1,576 offerings (77.60%), while Telugu represents 433 offerings (21.32%). Urdu accounts for 20 offerings (0.98%), and Hindi comprises 2 offerings (0.10%).



Institutional Breadth & Degree Availability Analysis

- **Institutional Degree Reach:** While B.Sc has the highest number of discrete course combinations, B.Com is present in 339 of 356 colleges (95.2%), confirming that commerce is the ubiquitous baseline for affiliated colleges. B.Sc is available in 293 colleges (82.3%), B.A in 147 colleges (41.3%), and BBA in 32 colleges (9.0%).
- **Course Breadth Dispersion:** The distribution of course offerings per institution is highly right-skewed. While the statistical mean is 5.7 offerings, the median institution offers exactly 5 courses. The bottom 25% of colleges offer 3 or fewer courses, whereas the upper tail features large autonomous colleges offering up to 27 courses.



B.Sc accounts for approximately 48.25% (980 of 2,031) of all parsed course offerings. This reflects extensive curriculum permutations across physical and life sciences.

Insight 2 - B.Com Possesses Broadest College Coverage

B.Com is offered by 95.2% of colleges (339 out of 356). Although B.Sc has more course combinations, B.Com is the foundational commercial track across nearly every affiliated campus.

Insight 3 - English is the Dominant Medium University-Wide

English represents 77.6% of all course offerings (1,576 courses), serving as the de facto standard of higher education instruction in scientific, technological, and corporate disciplines.

Insight 4 - B.A Exhibits an Inverted Regional Language Pattern

B.A shows 160 Telugu-medium offerings compared to only 94 in English. This is the single most significant sociolinguistic finding of the assessment, reflecting regional demand in public services.

Insight 5 - Course Availability is Highly Uneven Across Colleges

The median college offers 5 course programs, while the top institution offers 27. Over 75% of colleges offer 7 or fewer courses, showing that broad multidisciplinary portfolios are confined to premier urban colleges.

Insight 6 - Computing & General Tracks Form the University Backbone

General (369 offerings) and Computers (301 offerings) are the top two subcourse labels, followed by applied computing combinations like Math/Stats/CS (173) and Math/Physics/CS (165).

Evidence-Based Strategic Recommendations

All recommendations are derived strictly from quantitative findings, providing university administrators and academic planners with actionable interventions to enhance educational accessibility, curriculum relevance, and institutional equity.

Recommendation 1: Capacity & Quality Optimization for B.Sc & B.Com

Evidence: B.Sc and B.Com jointly constitute 85.28% of all undergraduate course offerings (1,732 of 2,031).

Action: University authorities should prioritize lab infrastructure audits, laboratory modernization, and teaching faculty allocation within these two degree categories. Standardizing curriculum tracks and introducing continuous industry assessment will have the maximum systemic impact on student outcomes.

Recommendation 2: Safeguard Regional Language Equity in Humanities

Evidence: B.A Telugu-medium offerings (160) substantially exceed English offerings (94), representing 59.9% of humanities capacity.

Action: Educational planners must not eliminate vernacular options under generic digitalization mandates. Vernacular humanities courses serve as vital entry points for rural and socially disadvantaged students targeting civil service and state competitive exams. Planners should provide high-quality digital textbooks in Telugu while introducing transitional English communication modules.

Recommendation 3: Portfolio Broadening & Interdisciplinary Clusters for Rural Colleges

Evidence: Over 50% of colleges offer 5 or fewer course tracks, severely limiting students' elective flexibility in non-urban districts.

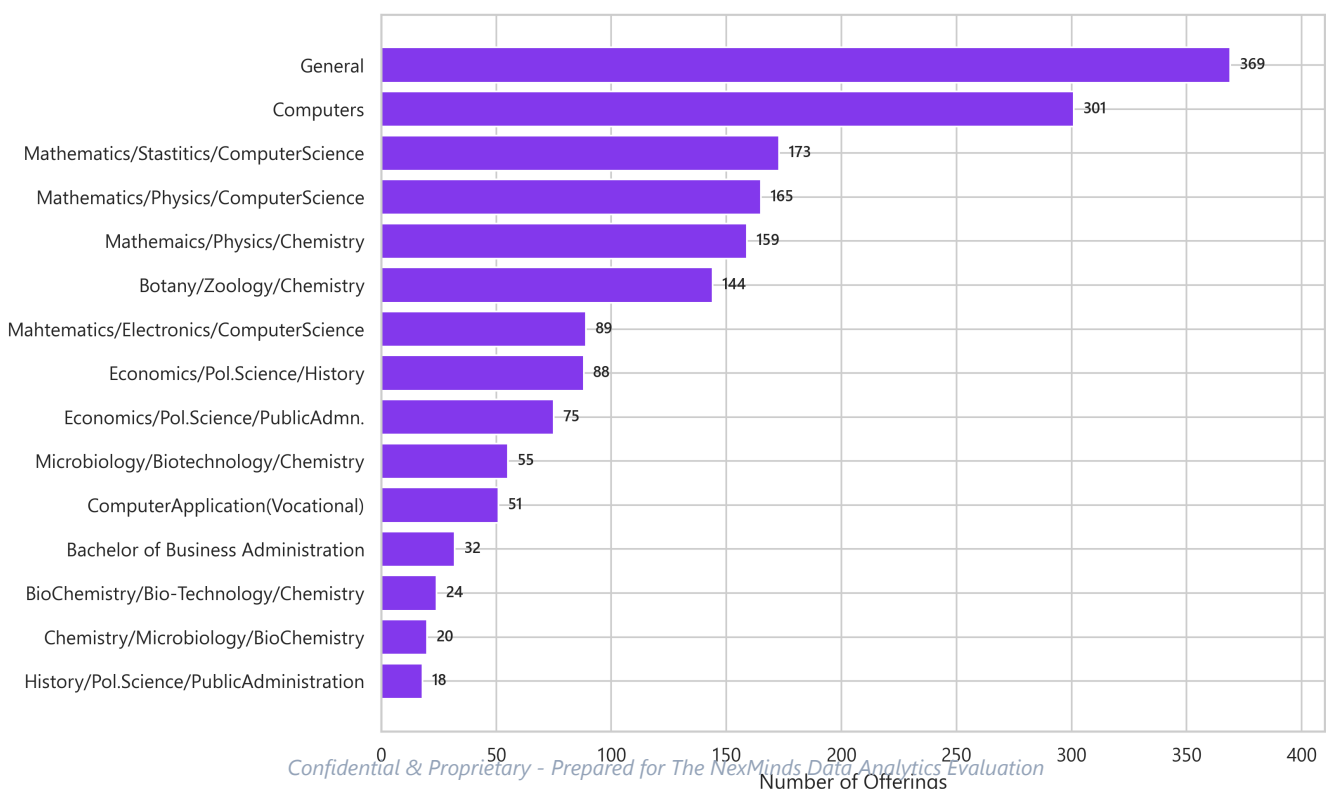
Action: Osmania University should implement regional hub-and-spoke academic networks where smaller colleges can share faculty for high-demand minors (e.g., Data Analytics, Financial Modeling, Biotechnology) through blended hybrid classrooms.

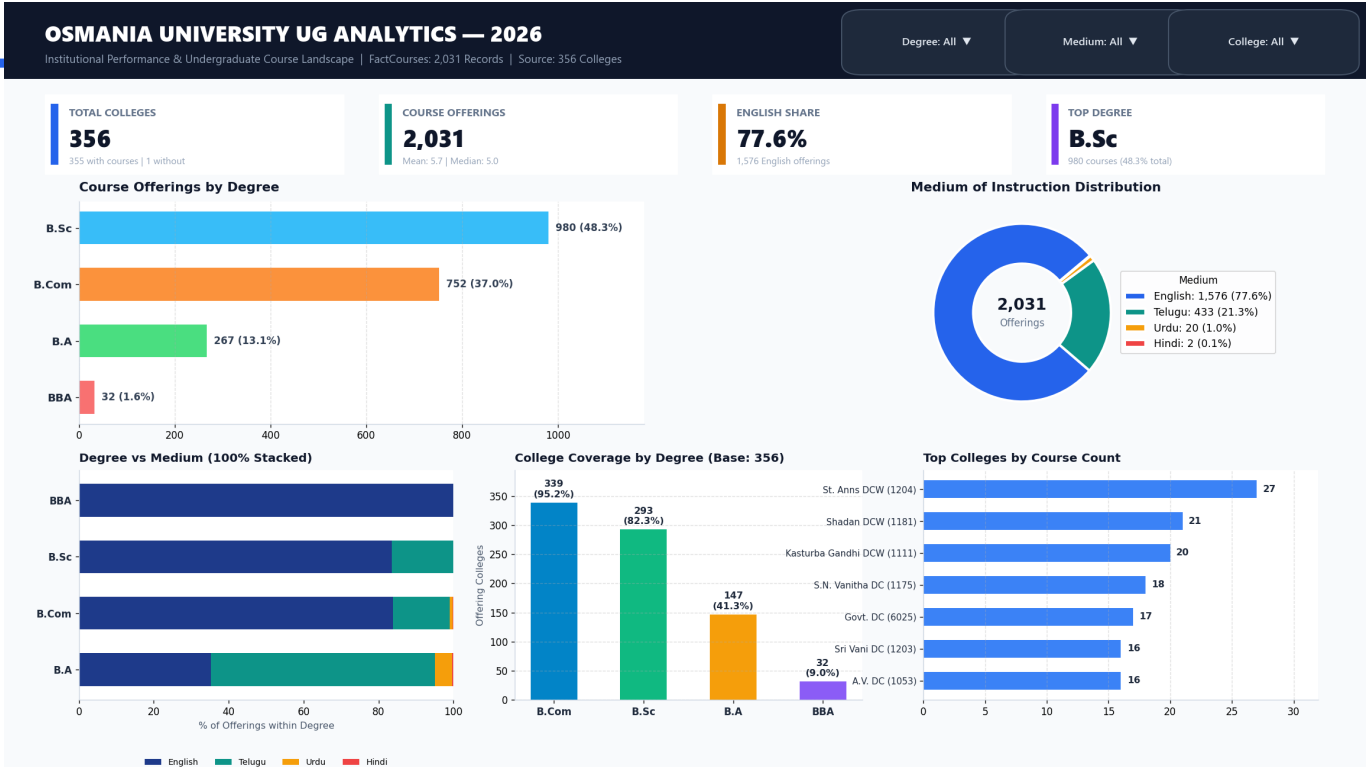
Recommendation 4: Industry Co-Certification of Applied Computing Tracks

Evidence: Computational programs (Computers, Math/Stats/CS, Math/Physics/CS, Vocational CA) account for over 700 aggregate offerings.

Action: Establish direct academic partnerships with IT/ITES industry leaders (NASSCOM, Telangana Academy for Skill and Knowledge) to validate and co-certify computing curricula, ensuring applied laboratory proficiency and direct employability upon graduation.

Top 15 SubCourse Combinations Across All Colleges





Analytical Conclusion

The analysis demonstrates a diverse but highly concentrated undergraduate course landscape across Osmania University's affiliated colleges. B.Sc and B.Com account for the vast majority of course offerings, while English represents the predominant operational medium. However, the B.A category reveals a distinctive Telugu-medium pattern, highlighting critical differences in language accessibility across disciplines. Academic breadth also varies considerably across colleges, with median institutions offering 5 courses while premier centers offer up to 27 distinct specializations.

Submission Deliverables Directory Structure

- The project is structured in strict conformance with the evaluation submission standard:
- README.md: Comprehensive architectural walkthrough, methodology, DAX formulas, and finding summaries.
 - data/: Contains original_dataset.xlsx, cleaned_course_data.xlsx, and dim_college.xlsx.
 - python/: NexMinds_Analysis.ipynb (fully executed Jupyter Notebook with 9 structured markdown sections).
 - powerbi/: NexMinds_UG_Analytics.pbix (interactive Star Schema report with DAX measures and slicers) + DAX_Measures.dax.
 - report/: NexMinds_Data_Analytics_Report.pdf (this executive report).
 - images/: High-resolution dashboard mockup (dashboard.png) and 7 standalone visual figures.

Assessment Completed & Verified By: Mohammed Farshan

Integrity Verified: FactCourses = 2,031 rows | DimCollege = 356 institutions | All DAX Formulas Validated